

A Fuzzy Nesterov-Accelerated Nonnegative Latent Factorization of Tensors Model for Efficient Representation to Dynamic Directed Graph Supplementary File

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I. INTRODUCTION

THIS is the supplementary file for the paper entitled “A Fuzzy Nesterov-Accelerated Nonnegative Latent Factorization of tensors Model for Efficient Representation to Dynamic Directed Graph”. We have put the proof of Lemma 1-2 and Theorem 1-2 in Section II, FNL algorithm in Section III, and the tables and figures of experimental results in Section IV.

II. MODEL CONVERGENCE ANALYSIS

The goal of the FNL model is to represent a HDI tensor $\mathbf{Y}$ via a nonnegative latent factorization. The optimization problem is defined as:

$$\min_{\mathbf{U}, \mathbf{W}, \mathbf{Z}} \varepsilon(\mathbf{U}, \mathbf{W}, \mathbf{Z}) = \sum_{(i,j,k) \in \Lambda} (y_{ijk} - \hat{y}_{ijk})^2 + \lambda^t (\sigma(u_{ir})^2 + \sigma(w_{jr})^2 + \sigma(z_{kr})^2), \quad (\text{S1})$$

where $\hat{y}_{ijk} = \sum_{r=1}^R \sigma(u_{ir}) \sigma(w_{jr}) \sigma(z_{kr})$, Λ and λ_t represent known element set and regularization coefficient on the t -th iteration, respectively, and $\sigma(*)$ represents Hard-Sigmoid function:

$$\sigma(\theta) = \max(0, \min(1, 0.2\theta + 0.5)), \quad (\text{S2})$$

where Λ and λ_t represent Known set and regularization coefficient on t -th iteration respectively.

For the convenience of analysis, we alternatively fix the counterpart of the active parameter as constant, i.e., we treat w_{jr} and z_{kr} as constant when performing the analyses with u_{ir} . The gradient of u_{ir} :

$$\nabla_{u_{ir} \varepsilon_{i,j,k}} = - \left(y_{ijk} - \sum_{r=1}^R \sigma(u_{ir}) \sigma(w_{jr}) \sigma(z_{kr}) \right) \sigma'(u_{ir}) \sigma(w_{jr}) \sigma(z_{kr}) + \lambda^t u_{ir}. \quad (\text{S3})$$

Note that the update of u_{ir} by NASGD is given as: For the t -th iteration counts.

$$\begin{aligned} t = 1 : u_{ir}^{t+1} &= u_{ir}^t - \eta^t (1 + \gamma^t) \nabla_{u_{ir} \varepsilon_{i,j,k}} (u_{ir}^t) \\ t \geq 2 : u_{ir}^{t+1} &= u_{ir}^t - \eta^t \nabla_{u_{ir} \varepsilon_{i,j,k}} (u_{ir}^t) \\ &\quad + \gamma^t ((u_{ir}^t - \eta^t \nabla_{u_{ir} \varepsilon_{i,j,k}} (u_{ir}^t)) - (u_{ir}^{t-1} - \eta^t \nabla_{u_{ir} \varepsilon_{i,j,k}} (u_{ir}^{t-1}))). \end{aligned} \quad (\text{S4})$$

It is worth noting that according to the fuzzy reasoning rule table (Table II), we can get:

$$\begin{aligned} \eta^t &\in [\eta_{min}, \eta_{max}], 0 < \eta_{min} \leq \eta^t \leq \eta_{max}, \\ \lambda^t &\in [\lambda_{min}, \lambda_{max}], 0 < \lambda_{min} \leq \lambda^t \leq \lambda_{max}, \\ \gamma^t &\in [\gamma_{min}, \gamma_{max}], 0 < \gamma_{min} \leq \gamma^t \leq \gamma_{max} < 1. \end{aligned} \quad (\text{S5})$$

In addition, in order to ensure the stability of NASGD iteration and the effective bounding of the error, we assume that the gradient function L^t satisfies Lipschitz continuity.

Assumption 1: since the function is convex, gradient satisfaction

$$\|\nabla_{u_{ir}\varepsilon_{i,j,k}^t}(u) - \nabla_{u_{ir}\varepsilon_{i,j,k}^t}(v)\| \leq L^t |u - v|, \quad (S6)$$

among them, u and v represent two different points on the gradient function L^t . Next calculate:

$$\begin{aligned} \nabla_{u_{ir}\varepsilon_{i,j,k}^t}(u) - \nabla_{u_{ir}\varepsilon_{i,j,k}^t}(v) &= q_{jr}t_{kr} [(y_{ijk} - \hat{y}_{ijk}(v))\sigma'(v) - (y_{ijk} - \hat{y}_{ijk}(u))\sigma'(u)] + \lambda^t(u - v) \\ &= q_{jr}t_{kr} ((y_{ijk} - \hat{y}_{ijk}(v))(\sigma'(v) - \sigma'(u)) + \sigma'(u)(\hat{y}_{ijk}(u) - \hat{y}_{ijk}(v))) + \lambda^t(u - v). \end{aligned} \quad (S7)$$

Since $\sigma(x) \leq 1$ and the maximum slope is 0.2, we have,

$$|\hat{y}_{ijk}(u) - \hat{y}_{ijk}(v)| = |\sigma(u) - \sigma(v)| q_{jr}t_{kr} \leq 0.2 |u - v| q_{jr}t_{kr}, \quad |\sigma'(u) - \sigma'(v)| \leq 0.2. \quad (S8)$$

By making $Q = \max_{j,r} q_{jr}t_{kr}$, then,

$$\|\nabla_{u_{ir}\varepsilon_{i,j,k}^t}(u) - \nabla_{u_{ir}\varepsilon_{i,j,k}^t}(v)\| \leq Q (|y_{ijk} - \hat{y}_{ijk}(v)| \cdot 0.2 + 0.2 + 0.2 |u - v| Q) + \lambda^t |u - v|. \quad (S9)$$

Set constant $M = \max |y_{ijk}|$, we can get,

$$|\hat{y}_{ijk}(u) - \hat{y}_{ijk}(v)| \leq M + Q. \quad (S10)$$

In summary,

$$\|\nabla_{u_{ir}\varepsilon_{i,j,k}^t}(u) - \nabla_{u_{ir}\varepsilon_{i,j,k}^t}(v)\| \leq L^t |u - v| = 0.04Q^2 + \lambda^t. \quad (S11)$$

Since L^t is kept bounded by the bounded derivatives of the hard_Sigmoid and the stability of the regularization term, the assumption that the gradient satisfies Lipschitz continuity holds. We define ω^t as:

$$\omega^t = \begin{cases} \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} + \eta^t \nabla_{u_{ir}\varepsilon_{i,j,k}^t}(u_{ir}^{t-1})), & t \geq 1 \\ 0, & t = 0 \end{cases}, \quad (S12)$$

where $\eta^t, \lambda^t, \gamma^t$ are dynamically adjusted according to the fuzzy reasoning $I^{(t)} = 0.5(\text{RMSE} + \text{MAE})$, we present the following results. *Lemma 1:* $\forall t \geq 0$, the following equation stands:

$$\begin{aligned} u_{ir}^{t+1} + \omega^{t+1} &= u_{ir}^{t+1} + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} (u_{ir}^{t+1} - u_{ir}^t + \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t)) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \gamma^t ((u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t)) - (u_{ir}^{t-1} - \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1}))) \\ &\quad + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} (u_{ir}^{t+1} - u_{ir}^t + \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t)) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \gamma^t (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \\ &\quad + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} \gamma^t (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \left(\gamma^t + \frac{\gamma^{t+1}\gamma^t}{1-\gamma^{t+1}} \right) (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \quad (S13) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \gamma^t \left(1 + \frac{\gamma^{t+1}}{1-\gamma^{t+1}} \right) (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \gamma^t \frac{1-\gamma^{t+1}+\gamma^{t+1}}{1-\gamma^{t+1}} (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \frac{\gamma^t}{1-\gamma^{t+1}} (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \\ &= u_{ir}^t - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) \\ &= u_{ir}^t + \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} + \eta^{t-1} \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^{t-1})) - \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) - \frac{\gamma^t}{1-\gamma^t} \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t) \\ &= u_{ir}^t + \omega^t - \frac{\gamma^t}{1-\gamma^t} \eta^t \nabla_{u_{ir}\varepsilon_{ijk}}(u_{ir}^t). \end{aligned}$$

Lemma 2: Let $u_{ir}^{-1} = u_{ir}^0$, for any $t \geq 0, \exists u_{ir}^*$:

$$\begin{aligned}
& \|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 \\
&= \left\| u_{ir}^t + \omega^t - \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) - u_{ir}^* \right\|^2 \\
&= \left\| (u_{ir}^t + \omega^t - u_{ir}^*) - \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\|^2 \\
&= (u_{ir}^t + \omega^t - u_{ir}^*)^2 - 2 \left\langle (u_{ir}^t + \omega^t - u_{ir}^*) \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\rangle + \left(\frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right)^2 \\
&= \|u_{ir}^t + \omega^t - u_{ir}^*\|^2 - 2 \left\langle (u_{ir}^t + \omega^t - u_{ir}^*) \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\rangle + \left(\frac{\eta^t}{1-\gamma^t} \right)^2 \|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\|^2.
\end{aligned} \tag{S14}$$

Note that (S13) can yield the appearance of u_{ir} when $t = 0$. By setting $u_{ir}^{-1} = u_{ir}^0$ the above inequality still holds. Hence, Lemma 2 stands.

Based on Lemmas 1-2, we present the following important result.

Theorem 1 (Convergence of NASGD): for any $t \geq 0$, let $\|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\| \leq G$, $|u_{ir}^{t+1} - u_{ir}^t| \leq A$, where G and A are both positive constants, by setting $B = A + 2\eta_{max}G$ as B be a positive constant, when $t \in \{0, \dots, T\}$, based on Lemma 2, we have:

$$\begin{aligned}
\|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 &= \|u_{ir}^t + \omega^t - u_{ir}^*\|^2 - 2 \left\langle (u_{ir}^t + \omega^t - u_{ir}^*) \frac{\eta^t}{1-\gamma^t} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \right\rangle \\
&\quad + \left(\frac{\eta^t}{1-\gamma^t} \right)^2 \|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\|^2.
\end{aligned} \tag{S15}$$

First of all, function $\varepsilon_{ijk}(u_{ir})$ is convex, therefore, about $(u_{ir}^t + \omega^t - u_{ir}^*) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)$, we have:

$$\begin{aligned}
& (u_{ir}^t + \omega^t - u_{ir}^*) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \\
&= (u_{ir}^t - u_{ir}^*) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) + \omega^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t) \\
&\geq \varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*) + \frac{\gamma^t}{1-\gamma^t} (u_{ir}^t - u_{ir}^{t-1} + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})) \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t).
\end{aligned} \tag{S16}$$

Secondly, since y_{ijk} , $\sigma(\cdot)$, λ_t are bounded, therefore, setting $\|\nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)\| \leq G$, we have:

$$\begin{aligned}
& |\omega^t \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^t)| \\
&\leq \frac{\gamma^t}{1-\gamma^t} G |u_{ir}^t - u_{ir}^{t-1} + \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})| \\
&\leq \frac{\gamma^t}{1-\gamma^t} G \gamma^{t-1} |u_{ir}^{t-1} - u_{ir}^{t-2} + \eta^{t-2} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-2}) - \eta^{t-1} \nabla_{u_{ir}} \varepsilon_{ijk}(u_{ir}^{t-1})| \\
&\leq \frac{\gamma_{max}}{1-\gamma_{max}} G (A + 2\eta_{max}G).
\end{aligned} \tag{S17}$$

Finally, substitute S(16) and S(17) into S(15) to get:

$$\begin{aligned}
\|u_{ir}^{t+1} + \omega^{t+1} - u_{ir}^*\|^2 &\leq \|u_{ir}^t + \omega^t - u_{ir}^*\|^2 - 2 \frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \\
&\quad + 2 \frac{\eta^t}{1-\gamma^t} \frac{\gamma_{max}}{1-\gamma_{max}} GB + \left(\frac{\eta^t}{1-\gamma^t} \right)^2 G^2.
\end{aligned} \tag{S18}$$

Accumulate from $t = 0$ to T :

$$\begin{aligned}
\|u_{ir}^{T+1} + \omega^{T+1} - u_{ir}^*\|^2 &\leq \|u_{ir}^0 + \omega^0 - u_{ir}^*\|^2 - 2 \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\
&\quad + 2 \frac{\gamma_{\max}}{1-\gamma_{\max}} GB \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} \right) + G^2 \sum_{t=0}^T \left(\left(\frac{\eta^t}{1-\gamma^t} \right)^2 \right).
\end{aligned} \tag{S19}$$

According to $\omega^0 = 0$, $\|u_{ir}^{t+1} + \omega_{ir}^{t+1} - u_{ir}^*\|^2 \geq 0$:

$$\begin{aligned}
&\sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\
&\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2} + \frac{\gamma_{\max}}{1-\gamma_{\max}} GB \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} \right) + \frac{1}{2} G^2 \sum_{t=0}^T \left(\left(\frac{\eta^t}{1-\gamma^t} \right)^2 \right) \\
&\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2} + \frac{\eta_{\max}}{1-\gamma_{\max}} \frac{\gamma_{\max}}{1-\gamma_{\max}} GB (T+1) + \frac{1}{2} G^2 \frac{\eta_{\max}}{1-\gamma_{\max}} (T+1).
\end{aligned} \tag{S20}$$

And then,

$$\begin{aligned}
&\frac{1}{T+1} \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\
&\leq \frac{\sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right)}{\frac{\eta_{\min}}{1-\gamma_{\max}} (T+1)} \\
&\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2 \frac{\eta_{\min}}{1-\gamma_{\max}} (T+1)} + \frac{\eta_{\max}}{1-\gamma_{\min}} \frac{\gamma_{\max}}{1-\gamma_{\max}} GB + \frac{1}{2} G^2 \frac{\eta_{\max}}{1-\gamma_{\max}}.
\end{aligned} \tag{S21}$$

Let $\bar{u}_{ir} = \sum_{t=0}^T u_{ir}^t / (T+1)$, from the convexity of the function, we can get:

$$\begin{aligned}
&\varepsilon_{ijk}(\bar{u}_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*) \\
&\leq \frac{1}{T+1} \sum_{t=0}^T \left(\frac{\eta^t}{1-\gamma^t} (\varepsilon_{ijk}(u_{ir}^t) - \varepsilon_{ijk}(u_{ir}^*)) \right) \\
&\leq \frac{\|u_{ir}^0 - u_{ir}^*\|^2}{2 \frac{\eta_{\min}}{1-\gamma_{\max}} (T+1)} + \frac{\eta_{\max}}{1-\gamma_{\min}} \frac{\gamma_{\max}}{1-\gamma_{\max}} GB + \frac{1}{2} G^2 \frac{\eta_{\max}}{1-\gamma_{\max}}.
\end{aligned} \tag{S22}$$

The proof of Theorem 1 can be done by fuzzy reasoning to ensure that η_t , λ_t , γ_t are obtained in a limited range. According to the same principle, $\varepsilon_{i,j,k}(w_{jr})$ converges by training w_{jr} by fixing u_{ir} and z_{kr} as a constant, and $\varepsilon_{ijk}(Z_{kr})$ converges by training z_{kr} by fixing u_{ir} and w_{jr} as a constant. Therefore, Theorem 1 stands.

III. FNL ALGORITHM

Algorithm I: FNL	
Input: Λ, I, J, K, R	
Operation	Cost
/* Initialization */	
1. Initialize $P, Q, T, V_{(U)}, V_{(W)}, V_{(Z)}$ Initialize U, W, Z at random in $(0, 0.05)$ Initialize $\eta, \lambda, \gamma, R = 10, T = \text{maximum round count}$	C_1
/* Training */	
/* Parameters Update Calculation */	
2. while not converge and $t \leq T$ do	
3. for each $y_{ijk} \in \Lambda$	
4. $\hat{y}_{ijk} = 0$	
5. for $r = 1$ to R	
6. Compute p_r, q_r, t_r and $\hat{y}_{ijk}$ based on (8), (9)	
7. end for	C_2
8. $err = y_{ijk} - \hat{y}_{ijk}$	
9. for $r = 1$ to R	
10. Update $U, W, Z, V_{(U)}, V_{(W)}, V_{(Z)}$ based on (18)	
11. end for	
12. end for	
/* Hyper-parameter Self-adaptation Calculation */	
13. Computing $\Delta(t)$ and $I^{(t)}$ based on (19)	
14. Computing deg1 and deg2 based on (20)	
15. Computing $\eta^{(t)}, \lambda^{(t)}, \gamma^{(t)}$ based on (21) and table S1	
16. $t = t + 1$	
17. end while	
/* Map Matrices Calculation */	
18. for $r = 1$ to R	
19. Map U, W, Z back to P, Q, T based on 8	C_3
20. end for	
/* Operation Ending */	
Output: P, Q, T built on Λ	

IV. EXPERIMENTAL RESULTS

TABLE S1
DETAILS OF FUZZY RULE

Δ	η	λ	γ
0.00001	2^{-7}	10^{-1}	0.8
0.0001	2^{-6}	10^{-2}	0.81
0.001	2^{-5}	10^{-3}	0.82
0.01	2^{-4}	10^{-4}	0.83
0.1	2^{-3}	10^{-5}	0.84
1	2^{-2}	10^{-6}	0.85

TABLE S2
HYPER-PARAMETER SETTINGS OF ALL MODELS

Datasets	Hyper-parameter Setting						
D1	M2: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.001$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.001$	M4: $R = 10$, $\lambda = 4 \times 10^{-8}$, $\gamma = 480$	M5: $R = 10$, $\lambda_1 = 0.04$, $\lambda_2 = 0.2$	M6: $R = 10$, $\lambda_1 = 0.5$, $\lambda_2 = 0.0625$	M7: $R = 10$, $\eta = 0.1$, $\lambda = 0.1$	M8: $R = 10$, $\eta = 0.1$, $\lambda = 0.001$
D2	M2: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.001$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.002$	M4: $R = 10$, $\lambda = 10^{-8}$, $\gamma = 500$	M5: $R = 10$, $\lambda_1 = 0.04$, $\lambda_2 = 0.2$	M6: $R = 10$, $\lambda_1 = 0.125$, $\lambda_2 = 0.25$	M7: $R = 10$, $\eta = 0.2$, $\lambda = 0.1$	M8: $R = 10$, $\eta = 0.1$, $\lambda = 0.0001$
D3	M2: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.001$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.004$	M4: $R = 10$, $\lambda = 10^{-8}$, $\gamma = 450$	M5: $R = 10$, $\lambda_1 = 0.005$, $\lambda_2 = 0.4$	M6: $R = 10$, $\lambda_1 = 0.25$, $\lambda_2 = 0.0625$	M7: $R = 10$, $\eta = 0.1$, $\lambda = 0.2$	M8: $R = 10$, $\eta = 0.2$, $\lambda = 0.01$
D4	M2: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.002$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.005$	M4: $R = 10$, $\lambda = 4 \times 10^{-8}$, $\gamma = 500$	M5: $R = 10$, $\lambda_1 = 0.007$, $\lambda_2 = 0.4$	M6: $R = 10$, $\lambda_1 = 0.5$, $\lambda_2 = 0.0625$	M7: $R = 10$, $\eta = 0.1$, $\lambda = 0.1$	M8: $R = 10$, $\eta = 0.2$, $\lambda = 0.001$
D5	M2: $R = 10$, $\lambda = 10^{-4}$, $\eta = 0.001$	M3: $R = 10$, $\lambda = 10^{-4}$, $\eta = 0.008$	M4: $R = 10$, $\lambda = 10^{-7}$, $\gamma = 480$	M5: $R = 10$, $\lambda_1 = 0.07$, $\lambda_2 = 0.3$	M6: $R = 10$, $\lambda_1 = 0.5$, $\lambda_2 = 0.0625$	M7: $R = 10$, $\eta = 0.2$, $\lambda = 0.2$	M8: $R = 10$, $\eta = 0.1$, $\lambda = 0.002$
D6	M2: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.001$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.004$	M4: $R = 10$, $\lambda = 10^{-8}$, $\gamma = 450$	M5: $R = 10$, $\lambda_1 = 0.006$, $\lambda_2 = 0.5$	M6: $R = 10$, $\lambda_1 = 0.25$, $\lambda_2 = 0.03125$	M7: $R = 10$, $\eta = 0.1$, $\lambda = 0.1$	M8: $R = 10$, $\eta = 0.2$, $\lambda = 0.01$
D7	M2: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.01$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.004$	M4: $R = 10$, $\lambda = 10^{-8}$, $\gamma = 480$	M5: $R = 10$, $\lambda_1 = 0.005$, $\lambda_2 = 0.4$	M6: $R = 10$, $\lambda_1 = 0.25$, $\lambda_2 = 0.0625$	M7: $R = 10$, $\eta = 0.1$, $\lambda = 0.2$	M8: $R = 10$, $\eta = 0.2$, $\lambda = 0.01$
D8	M2: $R = 10$, $\lambda = 10^{-4}$, $\eta = 0.01$	M3: $R = 10$, $\lambda = 10^{-5}$, $\eta = 0.004$	M4: $R = 10$, $\lambda = 10^{-8}$, $\gamma = 500$	M5: $R = 10$, $\lambda_1 = 0.004$, $\lambda_2 = 0.5$	M6: $R = 10$, $\lambda_1 = 0.5$, $\lambda_2 = 0.03125$	M7: $R = 10$, $\eta = 0.1$, $\lambda = 0.2$	M8: $R = 10$, $\eta = 0.1$, $\lambda = 0.01$

TABLE S3
RMSE, MAE, WIN/LOSS COUNTS OF M1-8 ON D1-8.

Datasets	M1	M2	M3	M4	M5	M6	M7	M8
D1	RMSE 0.3387 $\pm 6.8E-4$ MAE 0.2450 $\pm 1.8E-4$	0.3605 $\pm 8.3E-4$ 0.2498 $\pm 6.3E-4$	0.3431 $\pm 3.5E-4$ 0.2492 $\pm 2.1E-4$	0.3647 $\pm 7.1E-4$ 0.2557 $\pm 4.8E-4$	0.3498 $\pm 4.8E-5$ 0.2512 $\pm 7.4E-5$	0.3531 $\pm 4.5E-5$ 0.2508 $\pm 4.5E-5$	0.5871 $\pm 6.1E-5$ 0.4126 $\pm 7.6E-5$	0.5869 $\pm 4.2E-6$ 0.4111 $\pm 2.7E-6$
D2	RMSE 0.3466 $\pm 1.6E-5$ MAE 0.2474 $\pm 1.3E-5$	0.3697 $\pm 1.0E-3$ 0.2951 $\pm 6.4E-2$	0.3485 $\pm 5.2E-4$ 0.2552 $\pm 3.6E-4$	0.3675 $\pm 4.5E-4$ 0.2619 $\pm 4.5E-3$	0.3535 $\pm 7.4E-5$ 0.2556 $\pm 5.8E-5$	0.3534 $\pm 8.1E-5$ 0.2560 $\pm 4.9E-5$	0.6027 $\pm 1.4E-5$ 0.4259 $\pm 1.7E-5$	0.6024 $\pm 8.6E-6$ 0.4246 $\pm 1.1E-5$
D3	RMSE 0.3192 $\pm 3.9E-6$ MAE 0.2187 $\pm 2.1E-6$	0.3296 $\pm 2.2E-4$ 0.2234 $\pm 3.1E-4$	0.3252 $\pm 2.1E-0$ 0.2273 $\pm 2.0E-4$	0.3671 $\pm 2.4E-0$ 0.2471 $\pm 1.3E-3$	0.3344 $\pm 2.7E-5$ 0.2219 $\pm 8.1E-5$	0.3325 $\pm 1.8E-4$ 0.2282 $\pm 1.2E-4$	0.4758 $\pm 1.9E-6$ 0.3125 $\pm 4.3E-6$	0.4759 $\pm 3.9E-6$ 0.3123 $\pm 1.4E-6$
D4	RMSE 0.3161 $\pm 6.4E-6$ MAE 0.2166 $\pm 2.7E-6$	0.3359 $\pm 2.7E-4$ 0.2216 $\pm 5.6E-4$	0.3172 $\pm 1.0E-4$ 0.2186 $\pm 3.7E-5$	0.3855 $\pm 5.3E-4$ 0.2770 $\pm 3.2E-4$	0.3425 $\pm 7.5E-5$ 0.2314 $\pm 7.8E-5$	0.3489 $\pm 8.6E-5$ 0.2377 $\pm 1.6E-5$	0.4856 $\pm 9.0E-7$ 0.3242 $\pm 5.6E-6$	0.4857 $\pm 3.8E-6$ 0.3239 $\pm 1.5E-6$
D5	RMSE 0.2904 $\pm 2.1E-4$ MAE 0.1996 $\pm 1.8E-4$	0.3021 $\pm 5.5E-4$ 0.2033 $\pm 1.0E-3$	0.2939 $\pm 3.4E-4$ 0.2039 $\pm 2.8E-3$	0.2962 $\pm 8.5E-3$ 0.2029 $\pm 8.9E-3$	0.2985 $\pm 6.1E-5$ 0.2089 $\pm 2.0E-4$	0.3050 $\pm 9.6E-5$ 0.2161 $\pm 1.2E-5$	0.4794 $\pm 1.6E-6$ 0.3353 $\pm 4.7E-5$	0.4793 $\pm 1.8E-7$ 0.3336 $\pm 3.1E-6$
D6	RMSE 0.3426 $\pm 5.6E-6$ MAE 0.2493 $\pm 2.9E-6$	0.3717 $\pm 7.0E-4$ 0.2570 $\pm 2.4E-4$	0.3477 $\pm 2.6E-4$ 0.2555 $\pm 1.1E-4$	0.3489 $\pm 3.3E-4$ 0.2583 $\pm 2.1E-5$	0.3436 $\pm 5.4E-2$ 0.2599 $\pm 3.9E-4$	0.3440 $\pm 1.0E-5$ 0.2593 $\pm 5.1E-5$	0.5881 $\pm 5.0E-6$ 0.4295 $\pm 2.6E-5$	0.5880 $\pm 3.5E-6$ 0.4287 $\pm 7.0E-6$
D7	RMSE 0.3245 $\pm 2.2E-4$ MAE 0.2294 $\pm 3.8E-4$	0.3440 $\pm 8.3E-4$ 0.2420 $\pm 6.8E-4$	0.3294 $\pm 3.6E-4$ 0.2351 $\pm 4.4E-4$	0.3488 $\pm 7.3E-4$ 0.2478 $\pm 6.6E-4$	0.3359 $\pm 5.1E-5$ 0.2386 $\pm 5.8E-5$	0.3375 $\pm 6.4E-5$ 0.2397 $\pm 6.1E-5$	- -	- -
D8	RMSE 0.3117 $\pm 4.0E-4$ MAE 0.2164 $\pm 3.0E-4$	0.3304 $\pm 7.8E-4$ 0.2283 $\pm 6.0E-4$	0.3164 $\pm 4.2E-4$ 0.2283 $\pm 5.9E-4$	0.3251 $\pm 8.0E-4$ 0.2337 $\pm 6.2E-4$	0.3226 $\pm 6.2E-5$ 0.2251 $\pm 5.4E-5$	0.3242 $\pm 6.9E-5$ 0.2261 $\pm 5.6E-5$	- -	- -
Win/Loss	-	16/0	16/0	16/0	16/0	16/0	16/0	16/0
F-rank	1.00	4.28	2.59	4.88	3.63	4.31	7.75	7.31

TABLE S4
SINGLE TRAINING TIME COST IN RMSE/MAE(SECONDS), WIN/LOSS COUNTS OF M1-8 ON D1-8

Datasets		M1	M2	M3	M4	M5	M6	M7	M8
D1	RMSE	37.48 $\pm$ 0.66	1222.25 $\pm$ 2428.2	1358.66 $\pm$ 61.69	57.01 $\pm$ 5.65	180.48 $\pm$ 7.25	358.29 $\pm$ 6.03	1419.7 $\pm$ 113.98	522.58 $\pm$ 10.01
	MAE	39.82 $\pm$ 0.55	1318.37 $\pm$ 468.16	1357.62 $\pm$ 61.76	55.97 $\pm$ 3.42	92.55 $\pm$ 0.77	206.14 $\pm$ 2.85	1812.03 $\pm$ 132.77	1097.77 $\pm$ 54.89
D2	RMSE	22.74 $\pm$ 0.35	693.79 $\pm$ 40.64	700.9 $\pm$ 51.93	38.83 $\pm$ 2.82	79.12 $\pm$ 5.57	96.65 $\pm$ 3.98	986.4 $\pm$ 119.81	378.21 $\pm$ 3.17
	MAE	24.17 $\pm$ 2.35	740.65 $\pm$ 41.59	689.27 $\pm$ 44.32	38.83 $\pm$ 1.73	61.89 $\pm$ 3.34	54.67 $\pm$ 3.91	1222.48 $\pm$ 132.77	857.7 $\pm$ 2.31
D3	RMSE	4.99 $\pm$ 0.43	270.32 $\pm$ 95.46	122.41 $\pm$ 3.54	31.32 $\pm$ 3.03	25.05 $\pm$ 0.98	61.74 $\pm$ 4.18	394.79 $\pm$ 11.95	175.21 $\pm$ 2.63
	MAE	3.23 $\pm$ 0.32	272.08 $\pm$ 96.06	119.19 $\pm$ 4.27	30.5 $\pm$ 4.79	15.03 $\pm$ 1.11	51.38 $\pm$ 2.95	482.28 $\pm$ 7.46	342.87 $\pm$ 6.23
D4	RMSE	8.03 $\pm$ 0.55	171.97 $\pm$ 60.97	129.93 $\pm$ 0.28	22.63 $\pm$ 0.65	17.02 $\pm$ 0.63	34.27 $\pm$ 1.92	219.58 $\pm$ 5.08	158.55 $\pm$ 51.02
	MAE	4.59 $\pm$ 0.66	182.39 $\pm$ 69.95	128.18 $\pm$ 1.32	23.18 $\pm$ 0.62	12.07 $\pm$ 1.68	40.09 $\pm$ 1.72	275.74 $\pm$ 10.48	296.24 $\pm$ 96.35
D5	RMSE	8.88 $\pm$ 0.66	138.95 $\pm$ 9.59	128.5 $\pm$ 3.71	11.71 $\pm$ 0.94	26.98 $\pm$ 0.41	42.2 $\pm$ 3.53	390.7 $\pm$ 14.97	272 $\pm$ 16.51
	MAE	9.24 $\pm$ 0.43	141.5 $\pm$ 8.23	127.79 $\pm$ 4.95	12.66 $\pm$ 0.52	14.68 $\pm$ 0.61	59.06 $\pm$ 2.34	509.75 $\pm$ 23.17	901.16 $\pm$ 11.82
D6	RMSE	5.33 $\pm$ 0.04	214.96 $\pm$ 3.58	258.28 $\pm$ 4.71	19.57 $\pm$ 2.82	35.77 $\pm$ 1.1	65.38 $\pm$ 1.91	799.05 $\pm$ 9.98	146.72 $\pm$ 26.42
	MAE	5.33 $\pm$ 0.04	192.14 $\pm$ 4.36	220.03 $\pm$ 7.75	6.34 $\pm$ 0.93	8.82 $\pm$ 0.65	116.52 $\pm$ 3.96	1125.55 $\pm$ 5.81	575.61 $\pm$ 93.9
D7	RMSE	4722.89 $\pm$ 129.8	153965 $\pm$ 13366	171167 $\pm$ 3378	7175 $\pm$ 311	22549 $\pm$ 398	44765 $\pm$ 671	-	-
	MAE	5028.77 $\pm$ 425.6	171479 $\pm$ 23392	171614 $\pm$ 11662	7040 $\pm$ 233	14130 $\pm$ 926.34	25803 $\pm$ 2207	-	-
D8	RMSE	2083.52 $\pm$ 110	135316 $\pm$ 11325.60	139236 $\pm$ 2862	4646 $\pm$ 263	15175 $\pm$ 337	210273 $\pm$ 568	-	-
	MAE	3137.18 $\pm$ 129	135923 $\pm$ 13284	140292 $\pm$ 3358	5592 $\pm$ 308	13190 $\pm$ 395	25836 $\pm$ 667	-	-
Win/Loss	-	16/0	16/0	16/0	16/0	16/0	16/0	16/0	16/0
F-rank	1.00	6.00	5.38	2.25	2.81	3.94	7.75	6.25	

TABLE S5
WILCOXON SIGNED-RANKS TEST FOR TABLE S3

Comparison	R+	R-	p-value	Rank-biserial correlation
M1 vs M2	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M3	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M4	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M5	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M6	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M7	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M8	136	0	1.53 $\times$ 10 ⁻⁵	1.0

TABLE S6
WILCOXON SIGNED-RANKS TEST FOR TABLE S4

Comparison	R+	R-	p-value	Rank-biserial correlation
M1 vs M2	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M3	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M4	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M5	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M6	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M7	136	0	1.53 $\times$ 10 ⁻⁵	1.0
M1 vs M8	136	0	1.53 $\times$ 10 ⁻⁵	1.0

TABLE S7
THE PERFORMANCE OF FNL ON D1.1-D1.4

Dataset	Metrics	D1.1	D1.2	D1.3	D1.4
M1	RMSE	0.3642 $\pm 9.47\text{E-}4$	0.3571 $\pm 8.26\text{E-}4$	0.3493 $\pm 7.14\text{E-}4$	0.3418 $\pm 5.47\text{E-}4$
	*Per	0.012 $\pm 1.13\text{E-}3$	0.94 $\pm 4.32\text{E-}2$	5.91 ± 0.92	27.84 ± 2.13
	MAE	0.2783 $\pm 3.14\text{E-}4$	0.2724 $\pm 2.68\text{E-}4$	0.2661 $\pm 2.23\text{E-}4$	0.2486 $\pm 1.92\text{E-}4$
M2	**Per	0.01 $\pm 1.22\text{E-}3$	0.89 $\pm 5.66\text{E-}2$	5.57 ± 0.53	26.26 ± 5.22
	RMSE	0.3928 $\pm 1.23\text{E-}3$	0.3846 $\pm 1.04\text{E-}3$	0.3752 $\pm 9.16\text{E-}4$	0.3690 $\pm 8.14\text{E-}4$
	*Per	1.25 $\pm 2.15\text{E-}2$	25.67 ± 5.77	193.42 ± 43.5	1176.24 ± 376
M3	MAE	0.2987 $\pm 9.83\text{E-}4$	0.2914 $\pm 8.47\text{E-}4$	0.2836 $\pm 7.65\text{E-}4$	0.2652 $\pm 6.83\text{E-}4$
	**Per	1.18 $\pm 1.91\text{E-}3$	24.21 ± 4.23	188.09 ± 11.9	1154.64 ± 355
	RMSE	0.3714 $\pm 6.23\text{E-}4$	0.3637 $\pm 5.14\text{E-}4$	0.3561 $\pm 3.96\text{E-}4$	0.3482 $\pm 3.47\text{E-}4$
M4	*Per	0.39 $\pm 5.98\text{E-}2$	8.92 ± 0.29	139.61 ± 47.4	1594.72 ± 288
	MAE	0.2836 $\pm 5.14\text{E-}4$	0.2768 $\pm 4.13\text{E-}4$	0.2697 $\pm 2.47\text{E-}4$	0.2574 $\pm 2.19\text{E-}4$
	**Per	0.37 $\pm 9.43\text{E-}2$	8.39 ± 0.65	137.23 ± 12.5	1499.03 ± 250
M5	RMSE	0.3953 $\pm 9.74\text{E-}4$	0.3872 $\pm 8.76\text{E-}4$	0.3774 $\pm 7.89\text{E-}4$	0.3671 $\pm 7.14\text{E-}4$
	*Per	0.79 $\pm 4.28\text{E-}2$	15.97 ± 1.60	29.47 ± 3.33	42.68 ± 2.43
	MAE	0.3012 $\pm 7.23\text{E-}4$	0.2938 $\pm 5.84\text{E-}4$	0.2857 $\pm 5.24\text{E-}4$	0.2714 $\pm 5.03\text{E-}4$
M6	**Per	0.75 $\pm 7.58\text{E-}2$	15.17 ± 1.54	26.49 ± 1.65	30.54 ± 2.31
	RMSE	0.3897 $\pm 7.24\text{E-}4$	0.3821 $\pm 6.37\text{E-}4$	0.3719 $\pm 5.63\text{E-}4$	0.3624 $\pm 5.26\text{E-}4$
	*Per	0.90 $\pm 3.95\text{E-}2$	18.94 ± 1.77	67.38 ± 6.73	172.91 ± 27.3
M7	MAE	0.2947 $\pm 9.76\text{E-}4$	0.2873 $\pm 8.63\text{E-}4$	0.2787 $\pm 8.14\text{E-}4$	0.2653 $\pm 7.84\text{E-}4$
	**Per	0.85 $\pm 5.32\text{E-}2$	17.88 ± 1.29	63.63 ± 7.38	157.66 ± 25.8
	RMSE	0.4043 $\pm 6.67\text{E-}4$	0.3967 $\pm 6.24\text{E-}4$	0.3864 $\pm 5.47\text{E-}4$	0.3768 $\pm 4.86\text{E-}4$
M8	*Per	0.95 $\pm 9.14\text{E-}2$	19.93 ± 1.92	71.26 ± 7.13	288.24 ± 38.8
	MAE	0.3083 $\pm 6.14\text{E-}4$	0.3008 $\pm 5.27\text{E-}4$	0.2923 $\pm 4.92\text{E-}4$	0.2787 $\pm 4.67\text{E-}4$
	**Per	0.90 $\pm 5.96\text{E-}3$	18.88 ± 1.19	67.29 ± 6.73	272.15 ± 92.2
M9	RMSE	0.6276 $\pm 8.47\text{E-}4$	0.6189 $\pm 7.86\text{E-}4$	0.6097 $\pm 7.26\text{E-}4$	0.5983 $\pm 6.67\text{E-}4$
	*Per	2.15 $\pm 1.98\text{E-}3$	34.67 ± 0.67	120.92 ± 21.42	1592.7 ± 159
	MAE	0.4554 $\pm 9.47\text{E-}4$	0.4472 $\pm 8.74\text{E-}4$	0.4391 $\pm 8.24\text{E-}4$	0.4251 $\pm 7.93\text{E-}4$
M10	**Per	2.14 $\pm 1.56\text{E-}3$	34.39 ± 0.39	119.67 ± 64.29	1497.1 ± 150
	RMSE	0.6221 $\pm 5.73\text{E-}4$	0.6139 $\pm 5.24\text{E-}4$	0.6043 $\pm 4.83\text{E-}4$	0.5927 $\pm 4.36\text{E-}4$
	*Per	2.16 $\pm 1.67\text{E-}3$	35.32 ± 5.37	186.93 ± 18.7	1783.2 ± 179
M11	MAE	0.4513 $\pm 3.67\text{E-}4$	0.4417 $\pm 3.34\text{E-}4$	0.4324 $\pm 3.04\text{E-}4$	0.4212 $\pm 2.76\text{E-}4$
	**Per	2.15 $\pm 1.57\text{E-}3$	35.01 ± 5.79	175.72 ± 17.23	1676.2 ± 111

*/**Time cost per training in RMSE/MAE (Seconds).

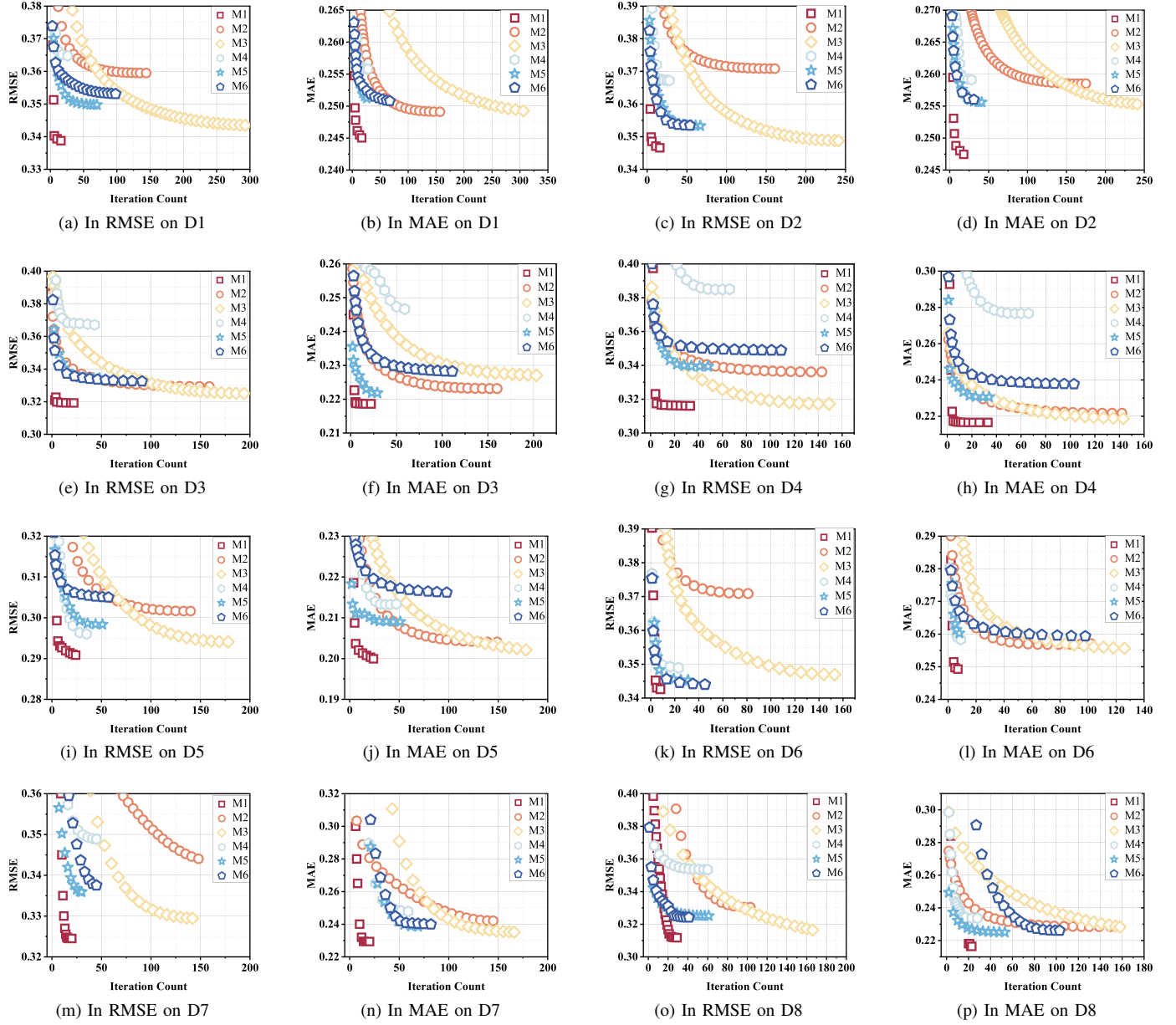


Fig. S1. Training curves of M1-6 in RMSE and MAE on D1-8.

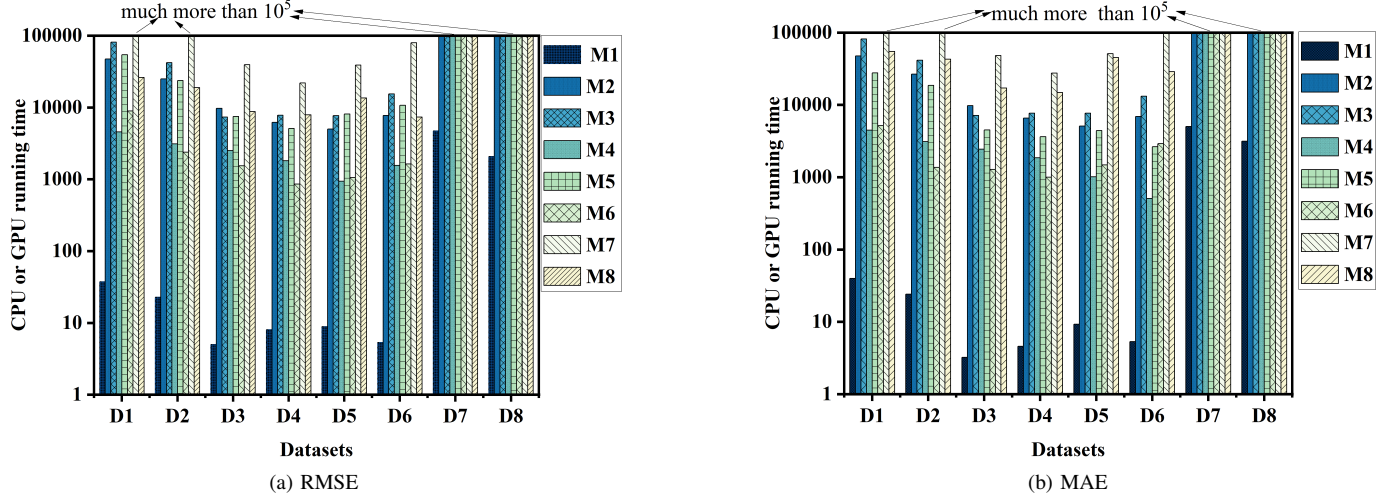
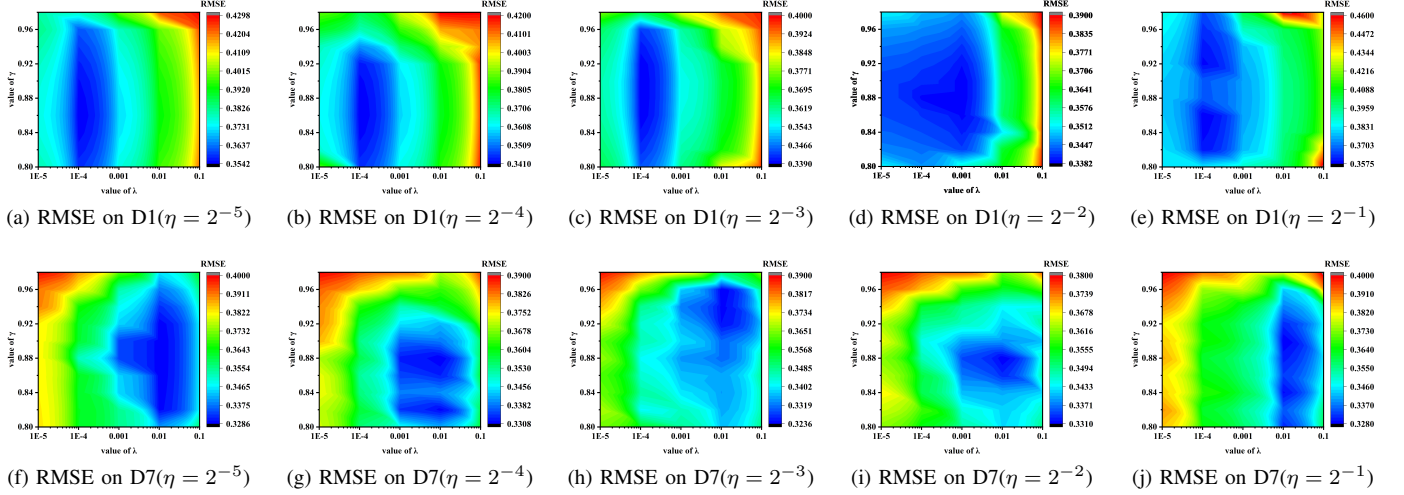
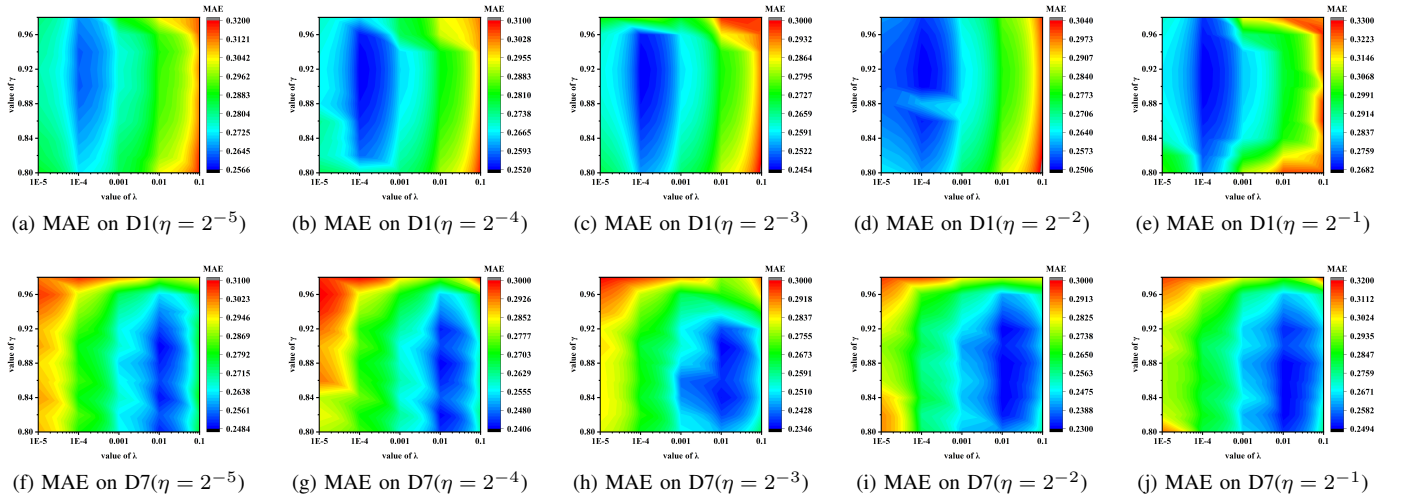


Fig. S2. The total time cost of M1-8 for obtaining the lowest RMSE and MAE on D1-8.

Fig. S3. RMSE of FNL as η , λ and γ varies on D1 and D7.Fig. S4. MAE of FNL as η , λ and γ varies on D1 and D7.

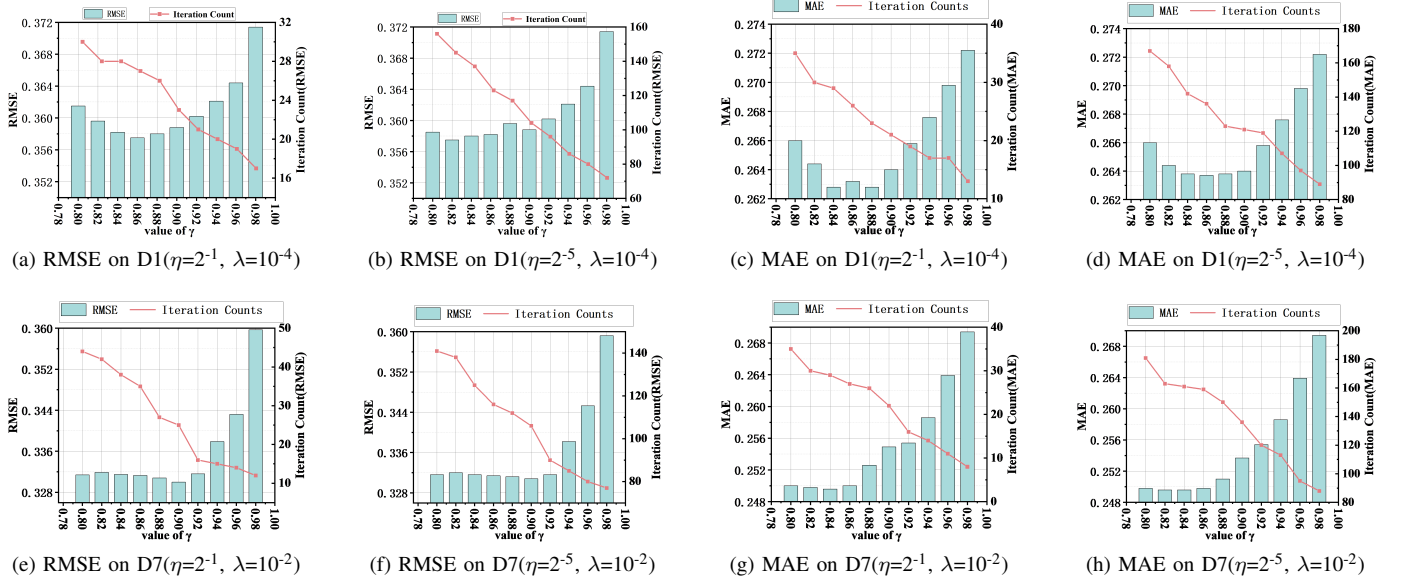
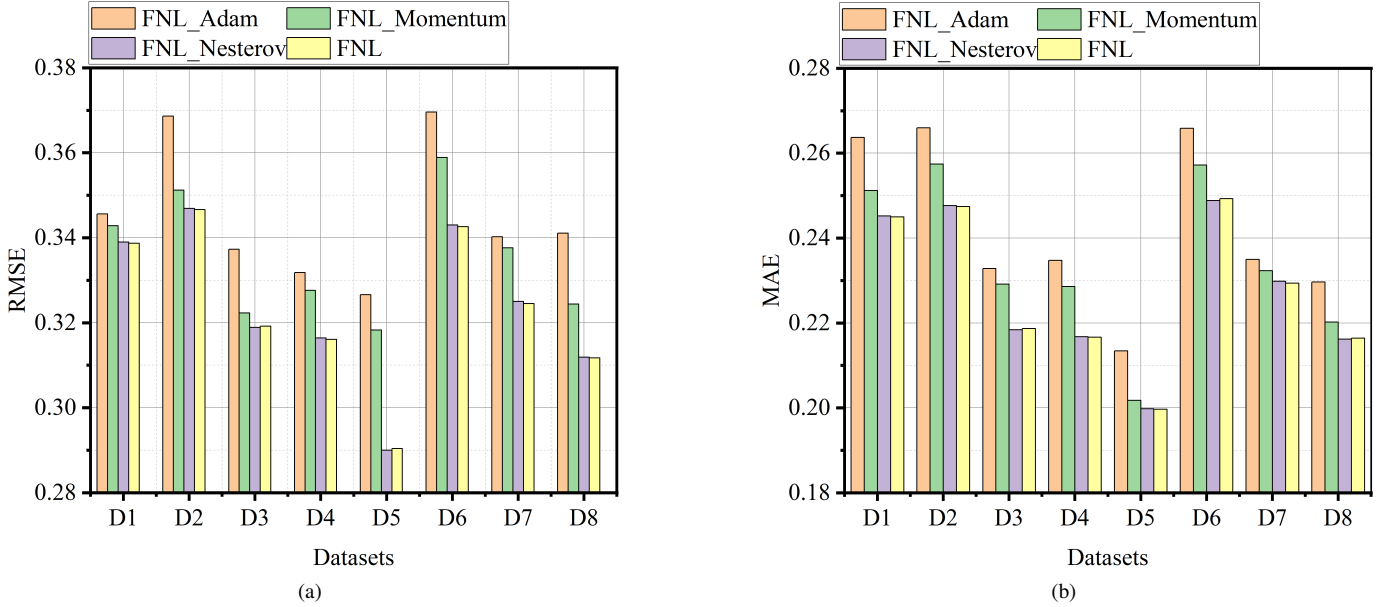
Fig. S5. RMSE/MAE and Iteration of FNL as η , λ and γ varies on D1 and D7.

Fig. S6. RMSE and MAE of FNL with different optimization methods on D1-D8.

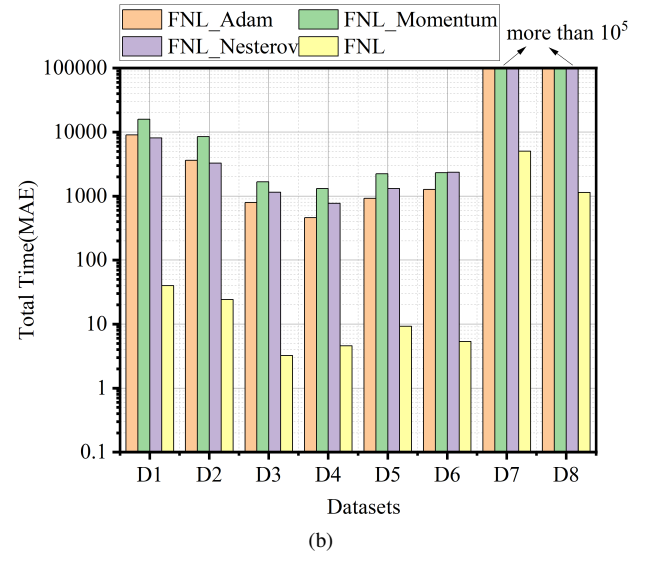
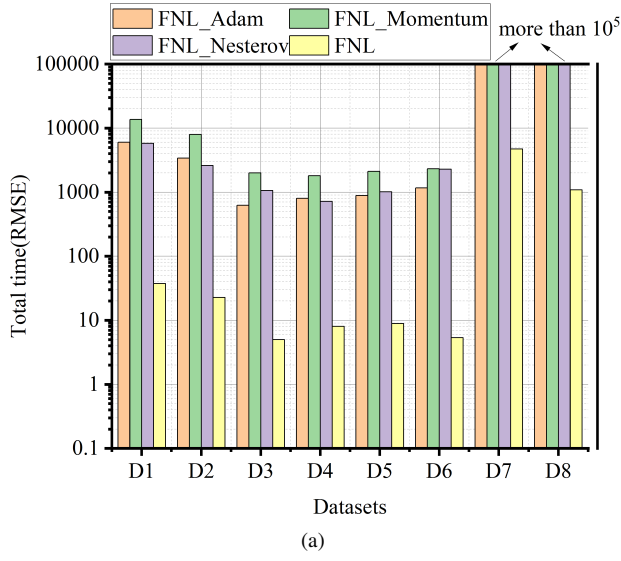
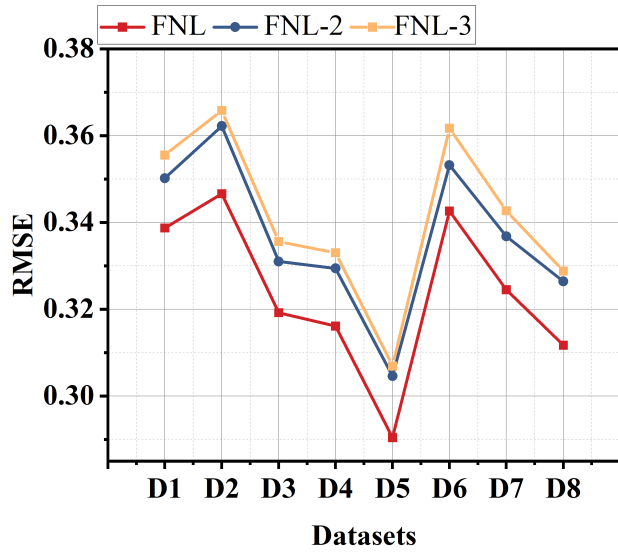
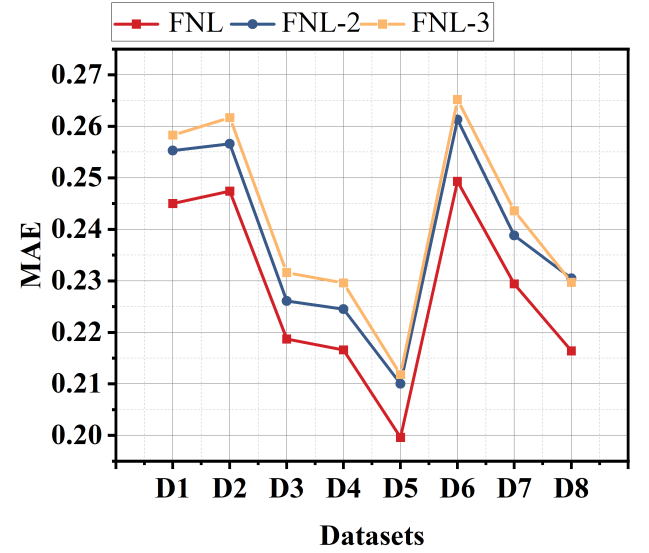


Fig. S7. The total time cost of FNL with different optimization methods for obtaining the lowest RMSE and MAE on D1-8.



(a) RMSE of FNL and FNL-2 on D1-D8



(b) MAE of FNL and FNL-2 on D1-D8

Fig. S8. Ablation study with nonlinear activation function on D1-D8.